PRODUCT & SERVICE CATALOG

Design | Development | Sheetmetal | Fabrication | 3D Printing

Engineering ideas, built to perform.

THE MECHLAB ENGINEERING

DESIGN • ENGINEER • DELIVER

Mechanical Design • CAD • Sheetmetal • Fabrication • 3D Printing

GET IN TOUCH

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About Us

Who we are and what we do

The MechLab Engineering

The MechLab Engineering is a mechanical engineering design studio that turns ideas into manufacturable products. We combine practical design experience with modern CAD and 3D printing tools to deliver parts and assemblies that are strong, efficient and cost-effective to produce. From product development and sheetmetal design to fabrication support, we cover the path from idea to finished part.

Whether you need a single bracket, a complete mechanism or a full set of production drawings, our team works closely with you from the first sketch to the final drawing package.

CAD — 3D modelling & assemblies

2D — Detailed drawings (GD&T)

3D — 3D printing services

SHEET — Sheetmetal design

FAB — Fabrication support

NPD — Product development

Our Services at a Glance

Mechanical Design

Concept design, mechanisms, machine parts, fixtures and jigs.

3D CAD Modelling

Parametric part and assembly models ready for manufacturing.

Product Development

End-to-end development from concept and prototype to a production-ready design.

Sheetmetal Designing

Sheetmetal parts, enclosures and brackets with flat patterns and bend details.

2D Drafting

Detailed production drawings with tolerances and GD&T.

Reverse Engineering

Recreate drawings and models from existing physical parts.

Fabrication Support

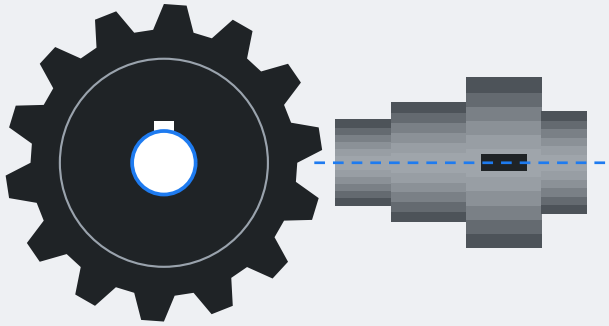
Drawings, bills of material and technical guidance for cutting, bending and welding.

3D Printing

Custom 3D-printed parts and prototypes in PLA, PETG, ABS and more.

Design & Engineering Services

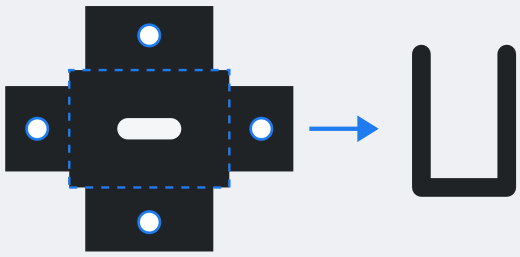
Detailed look at what we offer



3D CAD Modelling

Accurate parametric models of parts and assemblies with motion and interference checks, built for easy manufacture and later revisions.

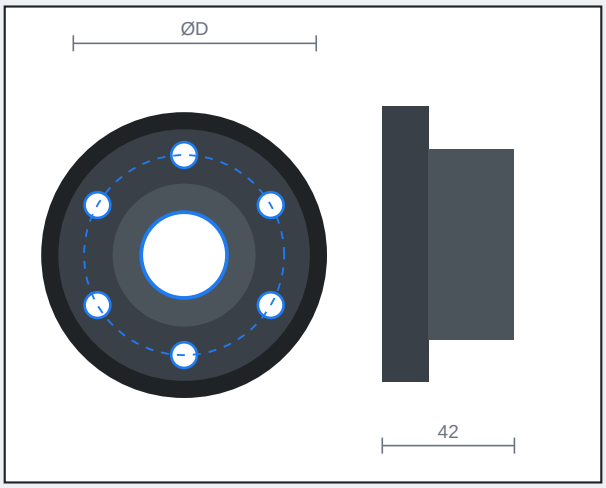
SolidWorks-style workflow



Sheetmetal Designing

Sheetmetal parts, enclosures and brackets with bend allowances, flat patterns and manufacturing-ready drawings.

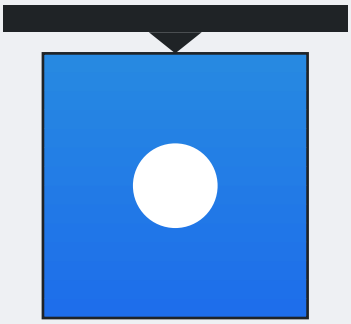
Sheetmetal



2D Drafting & GD&T

Clear manufacturing drawings with dimensions, tolerances, fits, surface finish and bill of materials.

Production-ready



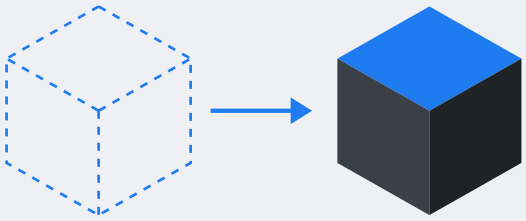
3D Printing & Prototyping

Custom 3D-printed prototypes and functional parts. Validate form, fit and function quickly before committing to tooling or machining.

3D printing

Product & Fabrication Services

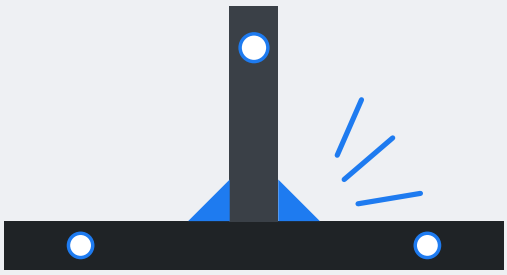
From first idea to finished part



Product Development

Take an idea from first sketch through concept, design, prototype and production-ready documentation.

Idea to product



Fabrication Support

Fabrication-ready drawings, bills of material and technical support for cutting, bending, welding and machining.

Manufacturing support

Let's build together.

Send us your requirement for a quick response.

EMAIL

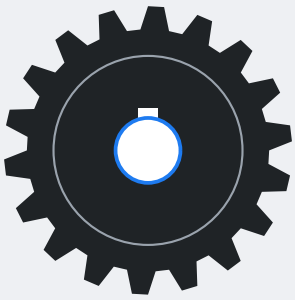
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Part Design Portfolio

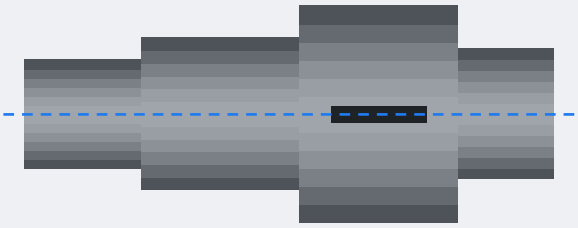
Typical components we design and detail



Spur Gear

Module-based gear design with bore, keyway and tooth-strength verification.

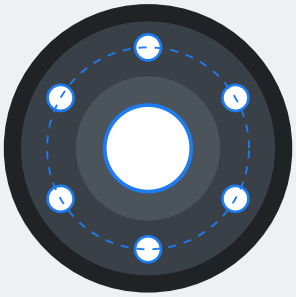
Power transmission



Stepped Shaft

Shafts with shoulders, keyways and bearing seats, sized for torque and bending.

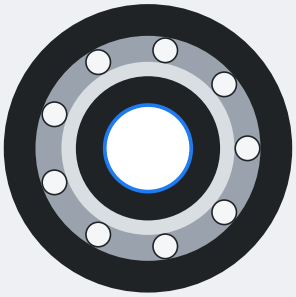
Rotating parts



Flange Coupling

Bolted flanges with PCD, bolt pattern and spigot features detailed for machining.

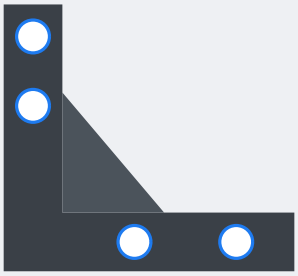
Connections



Bearing Assembly

Bearing housings and assemblies with correct fits, retention and sealing.

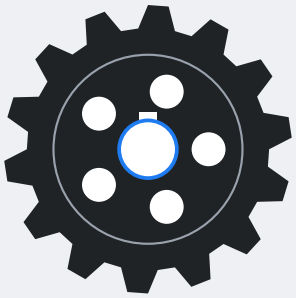
Supports



L-Bracket / Gusset

Structural brackets and mounts optimised for weight and stiffness.

Structures



Gear Plate

Lightened gears and plates with spoke-style cut-outs to save material.

Weight reduction

Illustrations are representative samples of component types. Custom designs are made to your specifications.

How We Work

Simple process, clear communication

01

Requirement

Share your idea, sketch or existing part and the working conditions.

02

Concept

We propose design options and discuss materials and manufacturing.

03

Modelling

Detailed 3D CAD, and interference checks to validate the design.

04

Drawings

Complete 2D drawings and files ready for fabrication.

05

Print & Support

Revisions, 3D-printed prototypes and production support.

Why Choose The MechLab Engineering

- Design-for-manufacture mindset - parts that are practical and economical to make
- Clear, well-documented drawings and models
- Fast turnaround and open communication
- Flexible support for one-off parts to complete assemblies
- Confidential handling of your designs

Let's build together.

Send us your requirement for a quick response.

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